

CoastNet: Cyclone Early-Warning & Coastal Relief Network

Introduction

Bangladesh's coastal belt is prone to seasonal cyclones, and the Ministry of Disaster Management has commissioned a unified digital network, CoastNet, to connect all agencies involved in cyclone early-warning and relief operations. You have been assigned to design and implement the base networks for seven operational units. The estimated number of hosts/agents at each unit is given below:

Unit	Abbreviation	Hosts
National Disaster Control Center	NDCC	280
Meteorological Radar Station	BMD	150
Coast Guard Base	CGB	200
Relief Logistics Hub	RLH	220
Field Rescue Camp	FRC	130
Telecom Relay Post	TRP	90
Remote Char (Island) Outpost	COI	40

Requirements

Topology Design & Server Setup

- Core Nodes: Every unit is represented by a router.
- Isolated Link: The Char Island Outpost (COI) is only connected to the Telecom Relay Post (TRP) — it has no other path into the network.
- Central Hub: NDCC, CGB, RLH, and TRP are all connected to one another via a shared central switch. Because a dedicated fibre line was laid during a past relief operation, NDCC and RLH also have an additional direct point-to-point link.
- The Warning Loop: A network loop connects the three agencies responsible for issuing warnings — NDCC is directly connected to BMD, BMD is connected to CGB, and CGB is connected back to NDCC.
- Single Spur: The Field Rescue Camp (FRC) is connected only to RLH.
- Each unit has its own LAN switch, exactly two PCs representing all hosts/agents, and one dedicated network printer.
- Local Email Servers: Every unit runs its own local email server for internal communication, using hostnames of the form mail.ndcc.gov, mail.bmd.gov, etc.
- Cross-Domain Email: In addition, NDCC (mail.ndcc.gov) and RLH (mail.rlh.gov) must be configured so that at least two user accounts on each server can send and receive email with the other domain in both directions (e.g. officer1@ndcc.gov ↔ logistics1@rlh.gov).
- Dedicated Web Servers: Only NDCC (www.ndcc.relief.gov) and RLH (www.rlh.relief.gov) have dedicated web servers. Configure them so every device across all seven units can open both sites by domain name.

- Central Directory: Only NDCC hosts the central DNS server. Configure it so every device in every unit uses it for domain name resolution.

Addressing

The base network address is derived from two members of your group using the rule below.

Step	Rule
1	Take the last 4 digits of the 1st group member's student ID and add the four digits together (digit-sum). If the result is a single digit, write it as a two-digit number by prefixing a 0. This becomes the first octet.
2	Take the last 2 digits of the 2nd group member's student ID as-is. This becomes the second octet.
3	The third and fourth octets are 0, and the prefix mask is /16.

Example: Suppose the 1st member's ID is 20211045. The last four digits are 1,0,4,5, which sum to 10 — first octet = 10. Suppose the 2nd member's ID ends in ...3022, so the last two digits are 22 — second octet = 22. The base network address is therefore 10.22.0.0/16.

If a group has only one member, use that member's ID for both steps.

Subnet this base network using VLSM, allocating address blocks in the following order: NDCC, CGB, RLH, TRP, BMD, FRC, COI, and finally all point-to-point / inter-router links. Show the complete VLSM tree in your submission.

All PCs receive their IP addresses dynamically. Servers, routers, and printers are all configured with static/manual addressing.

DHCP Configuration

- NDCC's router acts as the DHCP server for the three Warning Loop units — NDCC, BMD, and CGB.
- RLH and TRP each run their own dedicated local DHCP server for their own unit.
- COI has no DHCP server of its own and uses TRP's DHCP server, since its only path to anywhere is through TRP.
- FRC has no local DHCP server. Instead, FRC's router must be configured to relay DHCP requests to RLH's DHCP server using an IP helper address, since RLH's server is not on FRC's local segment.

Routing

- Dynamic Routing: NDCC, BMD, and CGB share their routing tables with one another using RIPv2 (the Warning Loop).
- Full Awareness via Redistribution: NDCC's router must be aware of every network in the topology, some learned statically and some dynamically. NDCC must redistribute its static routes into RIPv2 so that BMD and CGB automatically learn about the RLH, TRP, FRC, and COI networks without being configured with any static routes of their own.
- Default Gateway: COI can reach the rest of the network only through TRP; configure a default (static) route on COI's router.
- Static Mapping: RLH and TRP must each be configured with static routes to reach every external network — no default routes are permitted on these two routers.

- Exit-Interface Routing: FRC must reach every other network using exit-interface static routes via its link to RLH (the route statements must name only the outgoing interface, not a next-hop IP).
- Redundancy — Recursive Routing: If the direct NDCC–RLH link fails, RLH must reach NDCC's network through an alternate path via TRP, configured as a recursive static route (i.e. pointing to an intermediate router's address, not directly to an interface).
- Redundancy — Floating Static Route: TRP must also keep a floating static backup route to the NDCC network via CGB. Its administrative distance must be calculated as:

$$AD = (\text{sum of all digits of the 1st group member's student ID}) \times (\text{number of members in the group})$$

- This floating route must sit in TRP's routing table but only become active if the primary RIPv2-learned route toward NDCC fails.

Testing & Verification

- After configuration, every device across all seven units must be able to ping every other device.
- You must demonstrate, by shutting down the relevant primary interface(s), that RLH's recursive backup route and TRP's floating static route each take over correctly, and that traffic is restored end-to-end.
- You must demonstrate two-way email exchange between the ndcc.gov and rlh.gov domains, and confirm both www.ndcc.relief.gov and www.rlh.relief.gov open from a PC in every unit.

You are allowed to make any valid and necessary networking assumptions not explicitly stated above, provided they do not conflict with, remove, or simplify any requirement listed here.

Deliverables

- The network mentioned above should be implemented in Cisco Packet Tracer, with the necessary devices and full configuration.
- **In your topology, you must show the network address of each network by placing notes in Cisco Packet Tracer.**
- After completion, you should be able to test the conditions imposed, and all the devices should be able to ping one another.
- You will have to submit the following:
 - Work Distribution among the group members [Who did which part]
 - **As a member of the group, you must configure at least one of these 4 setups: DHCP/Email/Static Routing/Dynamic Routing.** Then, you can divide the rest of the work among yourselves as you see fit.
 - The pkt/pka file
 - Picture of the Network topology diagram with proper labels [You have to show the network addresses using notes for each network]
 - A PDF containing
 - VLSM tree
 - IP address table
 - The configuration commands of all the routers you have implemented.